A Practical Guide to Random forest

TOPIC -- RANDOM FOREST

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ExtraTrees Adding one further step of randomization yields extremely randomized trees, or ExtraTrees. As with ordinary random forests, they are an ensemble of individual trees, but there are two main differences: (1) each tree is trained using the whole learning sample (rather than a bootstrap sample), and (2) the top-down splitting is randomized: for each feature under consideration, a number of random cut-points are selected, instead of computing the locally optimal cut-point (based on, e.g., information gain or the Gini impurity). The values are chosen from a uniform distribution within the feature's empirical range (in the tree's training set). Then, of all the randomly chosen splits, the split that yields the highest score is chosen to split the node. Similar to ordinary random forests, the number of randomly selected features to be considered at each node can be specified. Default values for this parameter are p {\displaystyle {\sqrt {p}}} for classification and p {\displaystyle p} for regression, where p {\displaystyle p} is the number of features in the model.

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Using out-of-bag error as an estimate of the generalization error. Measuring variable importance through permutation. The report also offers the first theoretical result for random forests in the form of a bound on the generalization error which depends on the strength of the trees in the forest and their correlation.

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Consistency results Assume that Y = m (X) + ε {\displaystyle Y=m(\mathbf {X})+\varepsilon } , where ε {\displaystyle \varepsilon } is a centered Gaussian noise, independent of X {\displaystyle \mathbf {X} } , with finite variance σ 2 < ∞ {\displaystyle \sigma ^{2}<\infty } . Moreover, X {\displaystyle \mathbf {X} } is uniformly distributed on [0 , 1] d {\displaystyle [0,1]^{d}} and m {\displaystyle m} is Lipschitz. Scornet proved upper bounds on the rates of consistency for centered KeRF and uniform KeRF.

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Relation between infinite KeRF and infinite random forest When the number of trees M {\displaystyle M} goes to infinity, then we have infinite random forest and infinite KeRF. Their estimates are close if the number of observations in each cell is bounded:

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The number B of samples (equivalently, of trees) is a free parameter. Typically, a few hundred to several thousand trees are used, depending on the size and nature of the training set. B can be optimized using cross-validation, or by observing the out-of-bag error: the mean prediction error on each training sample xi, using only the trees that did not have xi in their bootstrap sample. The training and test error tend to level off after some number of trees have been fit.

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Relationship to nearest neighbors A relationship between random forests and the k-nearest neighbor algorithm (k-NN) was pointed out by Lin and Jeon in 2002. Both can be viewed as so-called weighted neighborhoods schemes. These are models built from a training set { (x i , y i) } i = 1 n {\displaystyle \{(x_{i},y_{i})\}_{i=1}^{n}} that make predictions y ^ {\displaystyle {\hat {y}}} for new points x' by looking at the "neighborhood" of the point, formalized by a weight function W: y ^ = ∑ i = 1 n W (x i , x ′) y i . {\displaystyle {\hat {y}}=\sum _{i=1}^{n}W(x_{i},x')\,y_{i}.} Here, W (x i , x ′) {\displaystyle W(x_{i},x')} is the non-negative weight of the i'th training point relative to the new point x' in the same tree. For any x', the weights for points x i {\displaystyle x_{i}} must sum to 1. Weight functions are as follows:

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K k u f (0 , x) = ∑ k 1 , … , k d , ∑ j = 1 d k j = k k ! k 1 ! … k d ! (1 d) k ∏ m = 1 d (1 − | x m | ∑ j = 0 k m − 1 (− ln ■ | x m |) j j !) for all x ∈ [0 , 1] d . {\displaystyle K_{k}^{uf}(\mathbf {0} ,\mathbf {x})=\sum _{k_{1},\ldots ,k_{d},\sum _{j=1}^{d}k_{j}=k}{\frac {k!}{k_{1}!\ldots k_{d}!}}\left({\frac {1}{d}}\right)^{k}\prod _{m=1}^{d}\left(1-|x_{m}|\sum _{j=0}^{k_{m}-1}{\frac {\left(-\ln |x_{m}|\right)^{j}}{j!}}\right){\text{ for all }}\mathbf {x} \in [0,1]^{d}.}

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Variants Instead of decision trees, linear models have been proposed and evaluated as base estimators in random forests, in particular multinomial logistic regression and naive Bayes classifiers. In cases that the relationship between the predictors and the target variable is linear, the base learners may have an equally high accuracy as the ensemble learner.

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From random forest to KeRF Given a training sample D n = { (X i , Y i) } i = 1 n {\displaystyle {\mathcal {D}}_{n}=\{(\mathbf {X} _{i},Y_{i})\}_{i=1}^{n}} of [0 , 1] p × R {\displaystyle [0,1]^{p}\times \mathbb {R} } -valued independent random variables distributed as the independent prototype pair (X , Y) {\displaystyle (\mathbf {X} ,Y)} , where E ■ [Y 2] < ∞ {\displaystyle \operatorname {E} [Y^{2}]<\infty } . We aim at predicting the response Y {\displaystyle Y} , associated with the random variable X {\displaystyle \mathbf {X} } , by estimating the regression function m (x) = E ■ [Y ■ X = x] {\displaystyle m(\mathbf {x})=\operatorname {E} [Y\mid \mathbf {X} =\mathbf {x}]} . A random regression forest is an ensemble of M {\displaystyle M} randomized regression trees. Denote m n (x , Θ j) {\displaystyle m_{n}(\mathbf {x} ,\mathbf {\Theta } _{j})} the predicted value at point x {\displaystyle \mathbf {x} } by the j {\displaystyle j} -th tree, where Θ 1 , … , Θ M {\displaystyle \mathbf {\Theta } _{1},\ldots ,\mathbf {\Theta } _{M}} are independent random variables, distributed as a generic random variable Θ {\displaystyle \mathbf {\Theta } } , independent of the sample D n {\displaystyle {\mathcal {D}}_{n}} . This random variable can be used to describe the randomness induced by node splitting and the sampling procedure for tree construction. The trees are combined to form the finite forest estimate m M , n (x , Θ 1 , … , Θ M) = 1 M ∑ j = 1 M m n (x , Θ j) {\displaystyle m_{M,n}(\mathbf {x} ,\Theta _{1},\ldots ,\Theta _{M})={\frac {1}{M}}\sum _{j=1}^{M}m_{n}(\mathbf {x} ,\Theta _{j})} . For regression trees, we have m n = ∑ i = 1 n Y i 1 X i ∈ A n (x , Θ j) N n (x , Θ j) {\displaystyle m_{n}=\sum _{i=1}^{n}{\frac {Y_{i}\mathbf {1} _{\mathbf {X} _{i}\in A_{n}(\mathbf {x} ,\Theta _{j})}}{N_{n}(\mathbf {x} ,\Theta _{j})}}} , where A n (x , Θ j) {\displaystyle A_{n}(\mathbf {x} ,\Theta _{j})} is the cell containing x {\displaystyle \mathbf {x} } , designed with randomness Θ j {\displaystyle \Theta _{j}} and dataset D n {\displaystyle {\mathcal {D}}_{n}} , and N n (x , Θ j) = ∑ i = 1 n 1 X i ∈ A n (x , Θ j) {\displaystyle N_{n}(\mathbf {x} ,\Theta _{j})=\sum _{i=1}^{n}\mathbf {1} _{\mathbf {X} _{i}\in A_{n}(\mathbf {x} ,\Theta _{j})}} . Thus random forest estimates satisfy, for all x ∈ [0 , 1] d {\displaystyle \mathbf {x} \in [0,1]^{d}} , m M , n (x , Θ 1 , … , Θ M) = 1 M ∑ j = 1 M (∑ i = 1 n Y i 1 X i ∈ A n (x , Θ j) N n (x , Θ j)) {\displaystyle m_{M,n}(\mathbf {x} ,\Theta _{1},\ldots ,\Theta _{M})={\frac {1}{M}}\sum _{j=1}^{M}\left(\sum _{i=1}^{n}{\frac {Y_{i}\mathbf {1} _{\mathbf {X} _{i}\in A_{n}(\mathbf {x} ,\Theta _{j})}}{N_{n}(\mathbf {x} ,\Theta _{j})}}\right)} . Random regression forest has two levels of averaging, first over the samples in the target cell of a tree, then over all trees. Thus the contributions of observations that are in cells with a high density of data points are smaller than that of observations which belong to less populated cells. In order to improve the random forest methods and compensate the misestimation, Scornet defined KeRF by